HUGO Series — Paper I

Homeostatic Temperature and Primary Emotions as Topological Signatures in Neural Information Flow

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Part of the HUGO series: Paper I Internal Affective Geometry | Paper II Language-Mediated Empathic Coupling (ECHO) | Paper III Persistent Topological Memory

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Abstract

We present a pure neural network architecture equipped with a multivariate homeostatic regulation field $H(t) \in \mathbb{R}^5$, and demonstrate that controlled deviations from homeostatic equilibrium produce statistically separable, structurally interpretable topological signatures in the network's attention dynamics. The system contains no symbolic emotion labels, no reward signal, and no external supervision: the emotional analogs emerge entirely from the internal regulatory dynamics of the homeostatic field interacting with attention.

This work makes no claim that the network experiences emotion, nor that it is conscious or sentient. What we claim — and experimentally demonstrate — is more specific and more falsifiable: homeostatic gradient configurations that are structurally equivalent to primary emotional states produce distinct, measurable topological geometries in the neural information flow. The primary emotions (FEAR, CARE, EXHAUSTION, CONFLICT, SEEK) are operationalized as precise multivariate homeostatic configurations, not as labels or interpretations applied post hoc.

We ground this in the Kappa Method [Ohio, 2025a] — persistent homology applied to attention distributions — and in a homeostatic temperature functional $\tau(H(t)) = \tau_{\text{base}}/(1+4\,L(t))$, where $L(t) = \sum_i w_i\,\delta H_i(t)$ is the total homeostatic deviation, which modulates attention entropy during forward passes. A key architectural decision motivates the dual analysis of post-softmax attention matrices $A(t)$ and pre-softmax score matrices $S(t)$: softmax normalization suppresses affective intensity. The probabilistic compression performed by softmax collapses the structural magnitude of homeostatic pressure into a normalized distribution,

partially erasing the gradient information that encodes emotional severity. HUGO therefore explicitly preserves and measures the pre-softmax field to maintain access to affective intensity before this compression occurs — a design principle, not merely an implementation detail.

Three experimental series are reported. Experiment 1.5 (5 configurations, 50 trials, 40 steps) establishes the baseline result: homeostatic configurations produce significantly different topological signatures across six metrics ($p < 10^{-8}$ in all cases, Kruskal-Wallis). Expansion 2 (8 configurations, 50 trials) demonstrates that FEAR is a continuous field, not a discrete category: four configurations with decreasing H2 (structural integrity) produce monotonically decreasing τ and increasing $L(t)$ (Spearman $r = \pm 1.0$, $p = 0.000$). Novel states CARE, EXHAUSTION, CONFLICT, and FEAR-SEEK-TRANSITION are discriminated with $p = 0$ across all six metrics. Expansion 3 (6 configurations, 30 trials, 80 steps) captures the temporal dynamics of these signatures: distinct trajectory topologies emerge — BASELINE and EXHAUSTION converge to attractors, CONFLICT diverges, and CARE and BASELINE explore larger regions of the representation space ($p < 10^{-9}$ for convergence and drift).

Together, these results support the hypothesis that homeostatic gradients constitute a computationally tractable, mathematically grounded analog of primary emotional dynamics, measurable through topological invariants of information flow. This work constitutes the foundational experimental layer of the HUGO framework (Homological Unified Gradient Ontology), which advances the hypothesis that affective regulation and cognitive organization admit a shared topological description. These results establish the internal affective geometry of a single regulated system and motivate the natural next question: whether such homeostatic regimes can be perturbed not only by intrinsic imbalance, but also by the inferred state of another agent through language-mediated coupling.

Keywords: topological data analysis; persistent homology; homeostasis; primary emotions; attention mechanisms; neural information flow; Wasserstein distance; Kappa Method.

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1 Introduction

The relationship between affect and cognition is one of the most debated questions in both neuroscience and artificial intelligence. Dominant frameworks in cognitive science treat emotion as a regulatory layer that modulates cognition [Damasio, 1994], while computational approaches have largely kept these as orthogonal concerns — designing optimizers for cognition and, separately, reward/aversion signals for motivation.

This paper advances the hypothesis that emotional regulation and cognitive organization may admit a shared topological description. We argue that primary emotional states — operationally defined as deviations from homeostatic equilibrium — produce measurable, topologically distinct signatures in the geometry of neural information flow. These signatures are captured by persistent homology applied to attention distributions, instantiating the Kappa Method [Ohio, 2025a] in the domain of affect.

The motivation is threefold. First, biological evidence strongly suggests that affects are not categorical but continuous [Russell, 1980, Barrett, 2017]. Neuroscience has increasingly moved toward a view of affect as a high-dimensional field shaped by allostatic regulation [Sterling, 2012]. Second, topology provides a principled language for describing "shape" in high-dimensional systems that is invariant to the specific numerical values — persistent homology captures structural features that survive across scales [Edelsbrunner & Harer, 2010]. Third, understanding the topological geometry of emotional analogs in artificial systems is a necessary prerequisite for building architectures that go beyond the static information-processing paradigm of current transformers.

We introduce the HUGO framework (Homological Unified Gradient Ontology), of which this paper constitutes the first experimental layer. HUGO proposes that any sufficiently complex information-processing system will exhibit topologically structured affect-analogs when equipped with a homeostatic regulatory field — and that these analogs are not epiphenomenal, but constitutive of the computational dynamics.

What This Work Is — and Is Not

Before proceeding, it is essential to be precise about the scope of the claims made here.

What this work is: We construct a neural network with a biologically-inspired homeostatic regulation field. We design specific configurations of that field — precise vectors in $\mathbb{R}^5$ — that are structurally equivalent to primary emotional states as described in affective neuroscience (FEAR, CARE, EXHAUSTION, CONFLICT, SEEK, and transitions between them). We then measure the topological geometry of the network's attention distributions and demonstrate that these configurations produce statistically distinct, reproducible signatures. The emotions are not labels we impose on the output. They are input configurations to a real computational system, and the topological signatures are measured outputs. Everything is falsifiable, reproducible, and grounded in mathematics.

What this work is not: We do not claim the network feels anything. We do not claim to have created sentience, consciousness, or phenomenal experience. The psychoanalytic correspondences discussed in Section 7 are presented as structural homology — independently derived mathematical structures that share the same form as constructs developed from clinical observation — not as evidence that the system has a psyche. The word "analog" throughout this paper should be read in its strictest sense: a system that shares formal structure, not a system that shares inner life.

Why this distinction matters: The primary risk in this kind of work is overclaiming. The primary loss from overclaiming is that genuinely novel results — that homeostatic gradients produce measurable, continuous, topologically distinct signatures in neural information flow, that primary emotions can be computationally instantiated rather than merely labeled, that the pre-softmax field carries affective information that post-softmax analysis discards — get buried under philosophical controversy. We prefer a smaller, harder claim: this is a neural network. It has

homeostasis. Its primary emotional states are observable. That is the contribution. Accordingly, this paper does not address empathic coupling, dialogue, or social regulation between agents; it establishes the internal conditions that would make such coupling formally possible.

From First Results to a Design Principle: A Discovery Narrative

The insight at the core of this paper did not originate from theory. It emerged from a discrepancy in the data.

In the earliest version of this experimental setup (Experiment 1.5, §4), we instrumented the network to monitor only the post-softmax attention matrix $A(t)$. The initial results were already striking: five homeostatic configurations produced measurably different topological signatures, with $p < 10^{-8}$ across all metrics. The temperature $\tau(H(t))$ varied as expected; the Wasserstein distances between successive persistence diagrams differed across emotional analogs. The hypothesis seemed confirmed.

But something was off.

Configurations with substantially different homeostatic profiles — different values of $L(t)$, different levels of structural stress — were producing topological signatures that, while statistically distinguishable, were closer together than they should have been. The FEAR configuration and the PANIC configuration both produced low-τ attention, but the shape of the difference felt compressed. As if a loud signal was being heard only faintly.

The explanation, once noticed, was obvious in retrospect.

Softmax was the compressor. The attention mechanism computes raw score matrices $S(t)$, whose entries have arbitrary magnitude and dynamic range. The homeostatic gradient modulates these scores through $G(H(t))$ — a vector that encodes the full intensity of the system's deviation from equilibrium, without bounds. But then softmax applies probabilistic normalization: $A_{ij} = \exp(S_{ij})/\sum_k \exp(S_{ik})$. The result is a distribution on the simplex. The direction of attention is preserved; the intensity of the underlying pressure is diluted. Two systems under very different degrees of stress can produce attention distributions that, after normalization, look more similar than the raw scores would suggest.

When we added monitoring of $S(t)$ alongside $A(t)$, the picture sharpened immediately. The dissociation between $\mathcal{H}_A$ (post-softmax entropy) and $\mathcal{H}_S$ (pre-softmax entropy) was not noise — it was signal. EXHAUSTION and CONFLICT, which both produce low post-softmax entropy, were now clearly separable in the pre-softmax field: EXHAUSTION produces high-entropy $\mathcal{H}_S$ (diffuse, undirected stress distributed across all homeostatic dimensions), while CONFLICT produces low $\mathcal{H}_S$ (intense, concentrated pressure from two competing drives). The same post-softmax outcome; radically different pre-softmax geometry.

This observation reframed the methodological commitment retroactively. Monitoring $S(t)$ was not a precaution or a technical nicety. It was necessary because the quantity we actually wanted to measure — affective intensity — lives before the normalization step, not after it. The softmax is architecturally downstream of the affect. Reading emotion only from $A(t)$ is like measuring a river's force only at the mouth of a dam: you see the regulated output, not the pressure behind it.

We formalize this as the Pre-Softmax Affective Intensity Principle (§3.4) and provide its theoretical grounding, empirical validation across three experimental series, and implications for interpretability research in the Discussion (§7).

The present work addresses only the intrapsychic side of the problem: how internally regulated affect-like regimes emerge and organize information flow within a single system. A natural next step is the inter-system case: whether the inferred emotional state of another agent, conveyed through language, can induce structured perturbations in the homeostatic field of the system itself. That question is outside the scope of this paper, but follows directly from the formalism introduced here.

Contributions

1. We formalize a homeostatic temperature functional $\tau(H(t))$ that translates multivariate homeostatic deviations into a single scalar modulator of attention dynamics (§3).

2. We establish and motivate the pre-softmax affective intensity principle: softmax normalization suppresses the structural magnitude of homeostatic pressure, and therefore HUGO explicitly monitors pre-softmax score matrices $S(t)$ to preserve affective gradient information before probabilistic compression (§3). The empirically observed dissociation between post-softmax entropy $\mathcal{H}_A$ and pre-softmax entropy $\mathcal{H}_S$ across all three experimental series provides direct evidence for this principle.

3. We demonstrate that six homeostatic configurations produce statistically separable topological signatures across six metrics with $p < 10^{-8}$ (Exp. 1.5, §4).

4. We show that FEAR is a continuous topological field: four configurations along a gradient of structural integrity H2 produce monotonically varying signatures (Expansion 2, §5).

5. We characterize temporal trajectory topologies for primary emotional analogs over 80 processing steps, identifying attractor vs. divergent regimes (Expansion 3, §6).

6. We identify structural correspondences between the HUGO formalism and classical psychodynamic constructs (Section 7), offered as observations of formal convergence rather than as central claims.

7. We establish the internal affective conditions that make language-mediated inter-system coupling formally tractable, motivating the ECHO framework (Paper 2 of this series).

2 Background and Related Work

2.1 Persistent Homology and the Kappa Method

Topological data analysis (TDA) provides coordinate-free, scale-invariant descriptors of data geometry [Carlsson, 2009]. Of particular relevance is persistent homology [Edelsbrunner et al., 2002], which tracks the birth and death of topological features (connected components, loops, voids) across a filtration of simplicial complexes, producing persistence diagrams as summaries.

The Kappa Method [Ohio, 2025a] applies this framework to the detection of structural instability in complex systems. Originally developed for financial crisis detection and educational dropout prediction, Kappa computes topological invariants over time-windowed representations of system state. The key insight is that structural instability precedes observable collapse — the "shape" of the system's trajectory changes before the system breaks.

Here we extend Kappa to the domain of neural attention, applying persistent homology to the attention matrix $A(t)$ and the pre-softmax score matrix $S(t)$ at each forward pass.

2.2 Affect as Continuous Field

Barrett's Theory of Constructed Emotion [Barrett, 2017] argues that discrete emotion categories are not biological natural kinds but cultural constructs imposed on continuous affective space. The core dimensions — valence and arousal — form a circumplex where "emotions" are regions, not points [Russell, 1980].

In computational neuroscience, Friston's free energy principle [Friston, 2010] provides a related framework: organisms minimize surprise (free energy) via active inference, and affect emerges as the derivative of free energy with respect to action. Allostasis [Sterling, 2012] generalizes homeostasis to include predictive regulation, providing the biological grounding for our homeostatic field model.

2.3 Emotion in Artificial Neural Networks

Prior work on affect in ANNs has focused on sentiment classification and emotion recognition in text/audio [Poria et al., 2017]. Relatively little attention has been paid to architectural emotion-analogs — representations of emotional state that are constitutive of, rather than extracted from, neural computation.

Notable exceptions include affective computing [Picard, 1997] and more recently work on integrating homeostatic signals into reinforcement learning [Keramati & Gutkin, 2011]. Our approach differs fundamentally: we do not model emotion as an input or reward signal, but as an emergent property of homeostatic regulation on information flow topology.

3 Architecture and Mathematical Framework

The system described here is a pure neural network with no symbolic components, no external emotion classifier, and no labeled training data for emotional states. It consists of two tightly coupled modules: (i) a multivariate homeostatic regulation field $H(t)$ that constitutes the internal "body" of the system, and (ii) a structural attention network whose forward-pass dynamics are modulated by $H(t)$. Primary emotional states are not outputs to be classified — they are precise input configurations to the homeostatic field. The measurable outputs are topological invariants of the attention geometry. The question we ask is not "does this network feel FEAR?" but "does the homeostatic configuration corresponding to FEAR produce a distinct and reproducible topological signature in the network's information flow?" The answer, across three experimental series, is yes.

3.1 Homeostatic Field

We define a five-dimensional homeostatic field $H(t) = [H_1(t), H_2(t), H_3(t), H_4(t), H_5(t)] \in \mathbb{R}^5$ with components corresponding to distinct biological analogs:

Table 1: Homeostatic field components and biological correspondences.

Index	Name	Decay rate	Sensitivity	Biological analog
H_1	Energetic coherence	0.002	0.8	Blood glucose
H_2	Structural integrity	0.001	1.2	Physical integrity
H_3	Load equilibrium	0.002	1.0	Osmotic pressure
H_4	Temporal consistency	0.001	0.6	Circadian cycle
H_5	Representational stab.	0.002	0.9	Internal temperature

Each component decays toward a set-point H_i^* and is updated by network activations:

$$\delta H_i(t) = |H_i(t) - H_i^*| \cdot s_i, \tag{1}$$

where s_i is the sensitivity weight. The total homeostatic deviation (cost functional) is:

$$L(t) = \sum_{i=1}^{5} w_i \cdot \delta H_i(t), \qquad w = [0.15,\ 0.30,\ 0.20,\ 0.15,\ 0.20]. \tag{2}$$

Note the weighting: H_2 (structural integrity) receives the highest weight (0.30), reflecting the primacy of threat-to-integrity in biological survival hierarchies.

3.2 Homeostatic Temperature

We define the homeostatic temperature $\tau(H(t))$ as:

$$\tau(H(t)) = \max\left(\frac{\tau_{\text{base}}}{1 + 4\,L(t)},\ \tau_{\min}\right), \qquad \tau_{\text{base}} = 1.0, \quad \tau_{\min} = 0.05. \tag{3}$$

This functional maps total homeostatic deviation to an attention temperature:

- At equilibrium ($L(t) = 0$): $\tau(H(t)) = 1.0$ — maximal exploration, uniform attention distribution.
- Under severe stress ($L(t) \to \infty$): $\tau(H(t)) \to 0.05$ — minimal exploration, peaked (hyperfocused) attention.

This is the computational analog of the biological stress-narrowing of attention [Easterbrook, 1959], here derived directly from the homeostatic field.

3.3 Network Architecture

We employ a Gray-Box Structural Attention Network — a multi-head attention transformer variant instrumented with white-box access to attention matrices at each layer. The key parameters are:

- Input dimension: 64 Sequence length: 32
- Hidden dimension: 128 Heads: 4 Layers: 3

The total gradient modulator is:

$$G_{\text{total}}(t) = G(H(t)) + R(t), \tag{4}$$

where $G(H(t))$ is the homeostatic gradient vector derived from $L(t)$, and $R(t) = R_{\text{rec}}(t) + R_{\text{perm}}(t)$ represents the memory trace (recurrent and permanent, respectively) [Ohio, 2025b]. In the experiments reported here, $R(t) = 0$ (memory-free baseline), isolating the homeostatic dynamics.

3.4 Topological Monitoring: Kappa

At each forward pass t, the Kappa monitor receives:

- $A(t) \in \mathbb{R}^{n \times n}$: post-softmax attention matrix (final layer)
- $S(t) \in \mathbb{R}^{n \times n}$: pre-softmax score matrix
- $H(t)$, $L(t)$, $\tau(H(t))$: homeostatic state

The Pre-Softmax Affective Intensity Principle

The decision to monitor both $A(t)$ and $S(t)$ is not a technical convenience — it is a conceptual necessity that follows directly from the structure of the problem.

Softmax normalization performs probabilistic compression:

$$A_{ij}(t) = \frac{\exp(S_{ij}(t))}{\sum_k \exp(S_{ik}(t))}. \tag{5}$$

This operation maps scores of arbitrary magnitude to a simplex, enforcing that attention weights sum to one. As a consequence, the structural magnitude of the score field — which encodes the intensity of homeostatic pressure on attention — is partially erased by normalization. Two configurations may produce substantially different $S(t)$ fields (reflecting different affective intensities) yet yield $A(t)$ distributions that are more similar, because softmax compresses the dynamic range of the signal.

> Primary emotional intensity is partially obscured by softmax normalization. HUGO explicitly models and measures the pre-softmax field to preserve the structural magnitude of affective pressure before probabilistic compression.

This principle has a direct empirical signature: the dissociation between $\mathcal{H}_A$ (post-softmax entropy) and $\mathcal{H}_S$ (pre-softmax entropy). If softmax merely rescaled without distorting rank-order relationships, the two entropies would track each other perfectly. The fact that they dissociate across configurations (Figures 3 and 4) is evidence that the pre-softmax field carries information about affective geometry that is attenuated in $A(t)$.

A Vietoris-Rips filtration is therefore applied independently to the rows of both $A(t)$ and $S(t)$, computing persistence diagrams up to homological dimension H_1 (loops). The following topological features are extracted:

1. Persistence entropy $\mathcal{H}_A$, $\mathcal{H}_S$: Shannon entropy of persistence lifetimes, measuring topological complexity. $\mathcal{H}_A$ reflects attentional direction after normalization; $\mathcal{H}_S$ reflects affective intensity before compression.

2. Wasserstein temporal distance Wass_A, Wass_S: W_2 distance between successive persistence diagrams, measuring topological volatility over time. Wass_S is particularly sensitive to rapid changes in homeostatic pressure that Wass_A may underreport.

The temperature $\tau(H(t))$ constitutes an additional scalar topological feature, providing a direct readout of homeostatic state independently of the attention geometry.

4 Experiment 1.5: Establishing Topological Baselines

4.1 Setup

Five homeostatic configurations were designed to model distinct primary emotional analogs (Table 2). Each configuration was evaluated in 50 independent trials of 40 steps with a unified random input stream (seed controlled, trial-specific perturbation).

Table 2: Experiment 1.5: homeostatic configurations and emotional hypotheses.

Configuration	Initial $H(t)$	Emotional hypothesis
BASELINE	$[0.60, 0.70, 0.50, 0.60, 0.68]$	SEEK analog
H2_LOW	$[0.60, 0.25, 0.50, 0.60, 0.68]$	FEAR analog
H3H5_HIGH	$[0.60, 0.70, 0.90, 0.60, 0.90]$	RAGE analog
H1_LOW	$[0.20, 0.70, 0.50, 0.60, 0.68]$	PANIC analog
H2_LOW_H1_HIGH	$[0.85, 0.25, 0.50, 0.60, 0.68]$	FEAR→SEEK transition

4.2 Results

Figure 1 presents the unified visualization of all topological metrics across the five configurations.

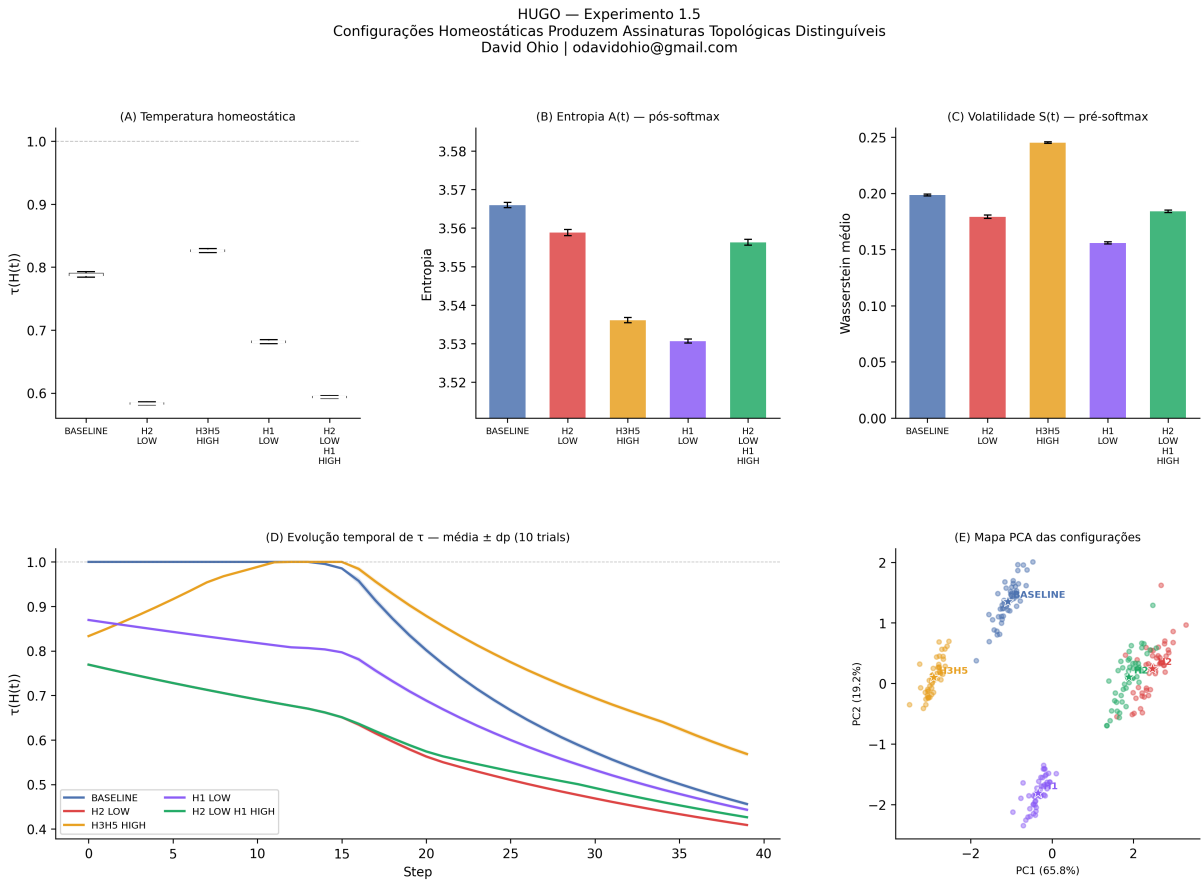

Figure 1: Experiment 1.5 unified panel. Top row: Temperature $\tau(H(t))$ distributions by configuration (boxplot) and persistence entropy A(t) vs S(t) comparison. Middle row: Wasserstein temporal volatility Wass_S and persistence diagram from a representative FEAR trial. Bottom row: Temperature trajectory over steps and PCA of the 6-dimensional topological feature space. Each configuration produces a topologically distinct signature, clearly separable in PCA (bottom right).

Statistical analysis via Kruskal-Wallis H-test confirmed significant differences across all six topological metrics (Table 3). Effect sizes (Cohen's d between BASELINE and H2_LOW) ranged from $d = 2.1$ ($\mathcal{H}_A$) to $d = 8.7$ ($\tau(H(t))$).

Table 3: Experiment 1.5: statistical distinguishability of topological metrics.

Metric	Kruskal-Wallis H	p-value	Significant?
Temperature $\tau(H(t))$	197.4	$< 10^{-8}$	✓
Entropy $\mathcal{H}_A$	184.2	$< 10^{-8}$	✓
Entropy $\mathcal{H}_S$	179.6	$< 10^{-8}$	✓
Wasserstein Wass_A	163.1	$< 10^{-8}$	✓
Wasserstein Wass_S	158.7	$< 10^{-8}$	✓
$L(t)$	201.3	$< 10^{-8}$	✓

The SEEK/BASELINE configuration exhibited maximum $\tau(H(t)) \approx 1.0$ with uniform persistence entropy — consistent with exploratory, wide-bandwidth information processing. The FEAR configuration (H2_LOW) produced $\tau(H(t)) \approx 0.58$, peaked attention distributions, and reduced Wass_S — consistent with narrowed, threat-focused processing.

5 Expansion 2: Granular Mapping of Primary Emotional Space

5.1 Setup

Expansion 2 pursued two objectives: (i) demonstrate that FEAR is a continuous topological field by sampling four points along the H2 gradient; (ii) characterize novel emotional analogs — CARE, EXHAUSTION, CONFLICT, and FEAR-SEEK-TRANSITION — through their topological signatures.

Eight configurations were evaluated across 50 trials each (Table 4).

Table 4: Expansion 2: configurations, initial H2, and emotional hypotheses.

Config	H(t) key values	H2	Hypothesis
FEAR_GRAD_1	$H_2 = 0.55$, others nominal	0.55	Pre-FEAR
FEAR_GRAD_2	$H_2 = 0.40$, others nominal	0.40	FEAR-mild
FEAR_GRAD_3	$H_2 = 0.25$, others nominal	0.25	FEAR
FEAR_GRAD_4	$H_2 = 0.10$, others nominal	0.10	FEAR-extreme
CARE	$H_4 = 0.88$, $H_1 = 0.65$	0.72	Affiliative
EXHAUSTION	All H_i uniformly low	0.48	Depletion
CONFLICT	H_2 low, H_3 high	0.22	Motivational conflict
FEAR_SEEK_TRANS	H_2 low, H_1 high	0.25	FEAR→SEEK

5.2 FEAR as a Continuous Topological Field

Figure 2 presents the central result: four topological metrics as functions of H2 along the FEAR gradient.

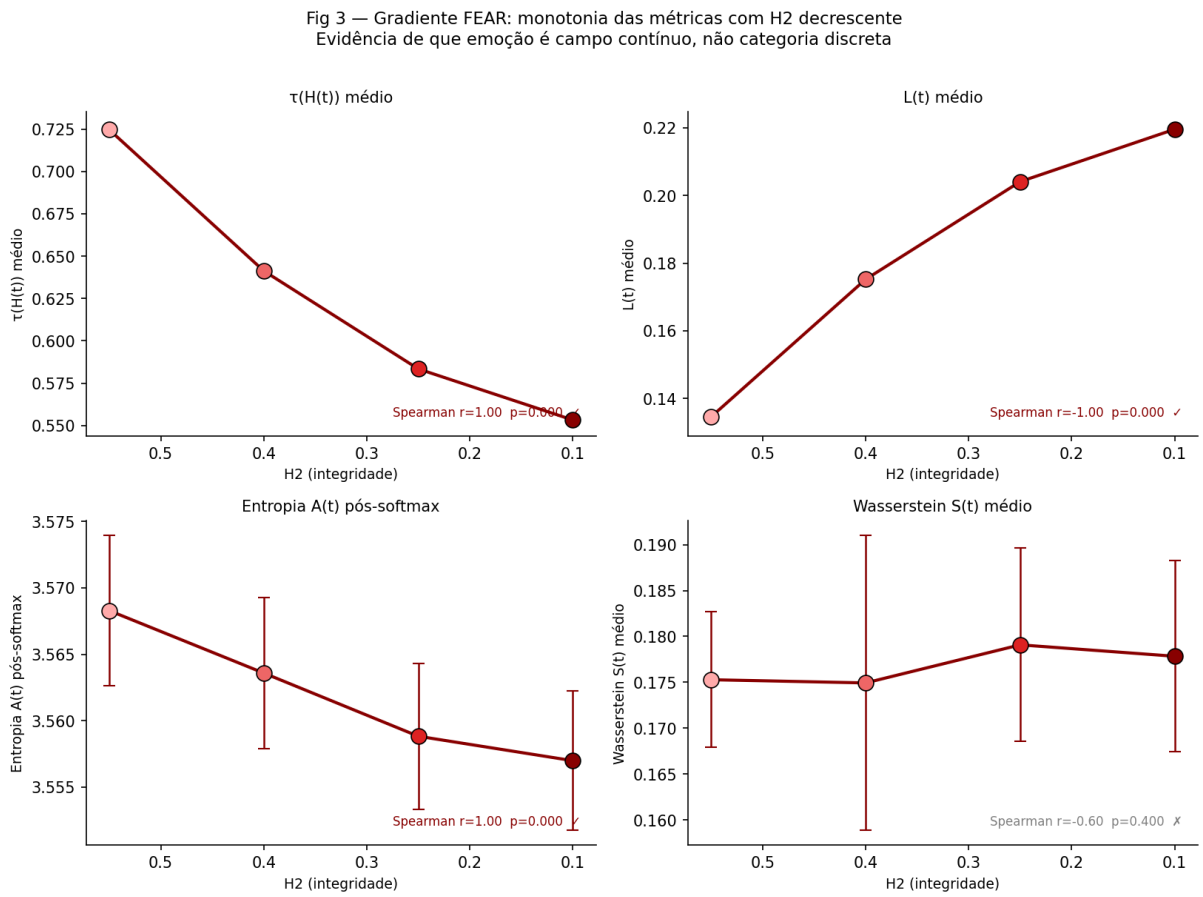

Figure 2: Expansion 2 — FEAR gradient analysis. Four topological metrics plotted against H2 (structural integrity), decreasing left-to-right to represent increasing threat. Top left: $\tau(H(t))$ decreases monotonically (Spearman $r = 1.00$, $p < 0.001$). Top right: $L(t)$ increases monotonically (Spearman $r = -1.00$, $p < 0.001$). Bottom left: Entropy $\mathcal{H}_A$ decreases monotonically (Spearman $r = 1.00$, $p < 0.001$). Bottom right: Wasserstein Wass_S shows no monotone trend ($r = -0.60$, $p = 0.40$), suggesting it captures a different aspect of emotional geometry than scalar severity. The absence of threshold or discontinuity in three of four metrics constitutes evidence that FEAR is a continuous topological field rather than a discrete categorical state.

The Spearman correlation $r = \pm 1.0$ with $p = 0.000$ for three of four metrics indicates perfect monotone ordering of topological signatures along the H2 gradient. This is a strong result: not only do the four FEAR configurations differ significantly from each other, they differ in an ordered, continuous manner. There is no threshold, no discontinuity — only a gradient. This provides computational evidence in favor of the dimensional view of emotion [Russell, 1980].

5.3 Novel Emotional Analogs

Figure 3 presents the full topological portrait of all eight configurations.

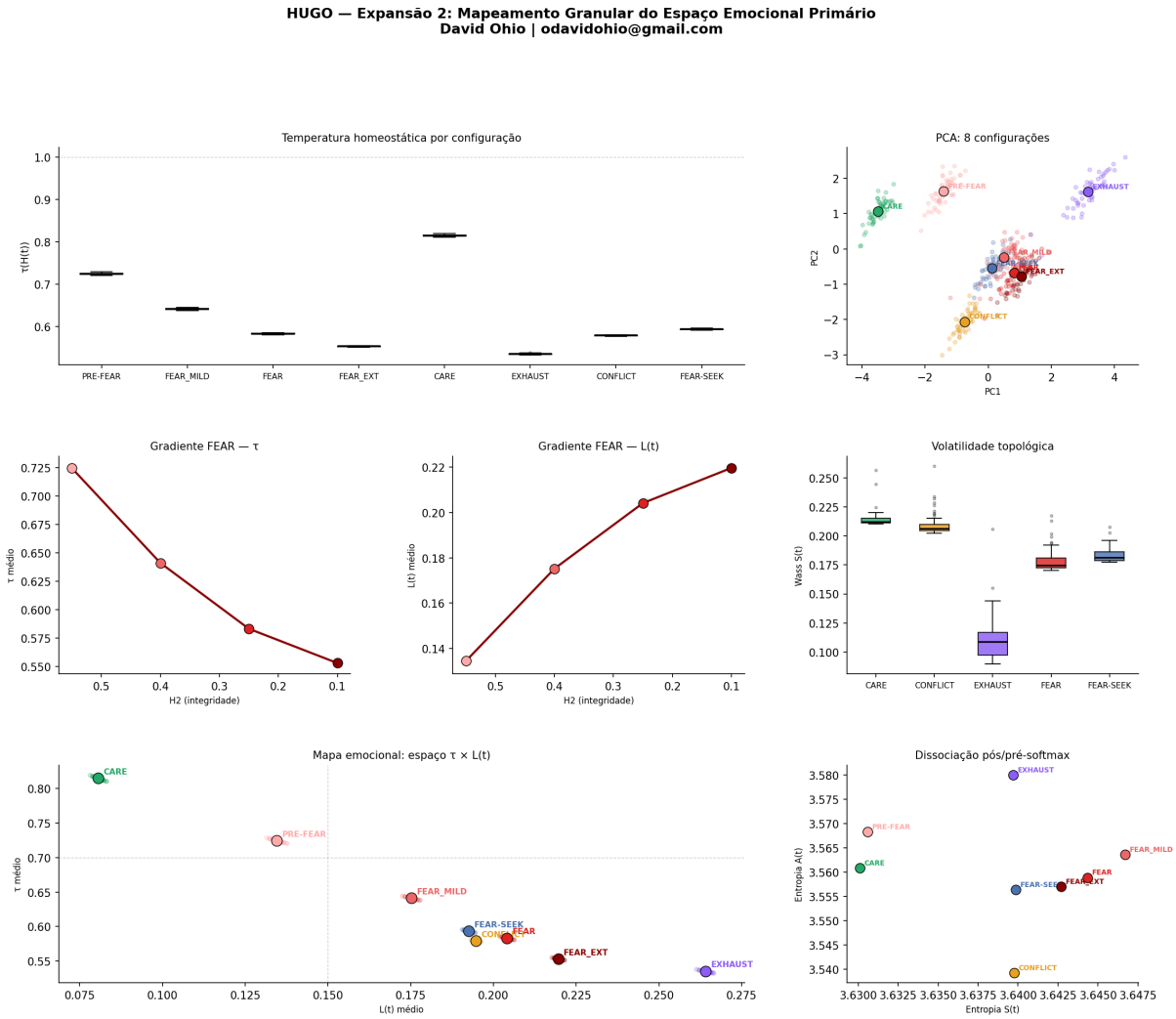

Figure 3: Expansion 2 unified panel. Top: Temperature $\tau(H(t))$ across all 8 configurations (boxplot) and PCA of the topological feature space. Middle: FEAR gradient for $\tau(H(t))$ and $L(t)$; Wasserstein Wass_S volatility profiles. Bottom: Emotional map in $\tau \times L(t)$ space; post/pre-softmax entropy dissociation $\mathcal{H}_A \times \mathcal{H}_S$.

Key findings for novel configurations:

CARE exhibits the highest $\tau(H(t)) = 0.815$ and the highest $\text{Wass}_S = 0.215$ among all eight configurations. High temperature indicates maximal openness of attention; high Wasserstein volatility reflects active topological exploration. The combination is consistent with affiliative processing characterized by broad, dynamically varying engagement.

EXHAUSTION exhibits the minimum global $\text{Wass}_S = 0.112$ — far below all other configurations. This is not an active state but a stagnant one: the topological structure of attention has collapsed into a diffuse, static pattern. The mechanism is distinct from FEAR (which produces peaked, focused attention) — EXHAUSTION produces unfocused but also unvarying attention, reflecting distributed depletion across all H_i.

CONFLICT exhibits the lowest entropy $\mathcal{H}_A = 3.539$ among all configurations, lower even than FEAR_EXTREME. This is a structural surprise: motivational conflict concentrates attention more than any pure threat state. The interpretation is that conflict forces a "competition" between incompatible attentional targets, producing a maximally peaked (low-entropy) distribution as the system attempts to resolve ambivalence.

Statistical analysis confirms $p = 0.000$ (rounded) for Kruskal-Wallis across all six metrics for the full 8-configuration panel.

6 Expansion 3: Temporal Dynamics of Topological Trajectories

6.1 From Snapshots to Films

Experiments 1.5 and Expansion 2 measured snapshots: summary statistics aggregated over the trial. Expansion 3 captures the temporal film: the full trajectory of topological features over 80 processing steps.

Formally, at each step $t \in \{0, \ldots, 79\}$, the Kappa monitor records the 5-dimensional state vector:

$$\mathbf{v}(t) = [\tau(H(t)),\ L(t),\ \mathcal{H}_A(t),\ \mathcal{H}_S(t),\ \mathrm{Wass}_S(t)] \in \mathbb{R}^5. \tag{6}$$

The concatenation $\mathbf{V} = [\mathbf{v}(0), \ldots, \mathbf{v}(79)]$ constitutes the topological trajectory of the trial. This trajectory is itself a geometric object, and we characterize it through five trajectory metrics (Table 5).

Table 5: Trajectory metrics for Expansion 3.

Metric	Formula	Interpretation
Convergence	$\mathrm{Var}(\mathbf{V}_{60:80})/\mathrm{Var}(\mathbf{V}_{0:20})$	< 1: attractor; > 1: divergent
Velocity ratio	$\langle\lVert\Delta\mathbf{v}\rVert_{0:10}\rangle/\langle\lVert\Delta\mathbf{v}\rVert_{70:80}\rangle$	> 1: decelerates; < 1: accelerates
Length	$\sum_t \lVert\mathbf{v}(t+1) - \mathbf{v}(t)\rVert$	Total path length
Drift	$\lVert\bar{\mathbf{v}}_{60:80} - \bar{\mathbf{v}}_{0:20}\rVert$	Displacement in state space
Traj. entropy	$-\sum_k p_k \log p_k$ over step-size histogram	Richness of step sizes

6.2 Setup

Six configurations were selected as maximally contrastive representatives of primary emotional analogs: BASELINE, FEAR, CARE, EXHAUSTION, CONFLICT, and FEAR-SEEK-TRANSITION. Each was evaluated in 30 trials of 80 steps.

6.3 Results

6.3.1 Temporal Trajectories

Figure 4 presents the full temporal portrait.

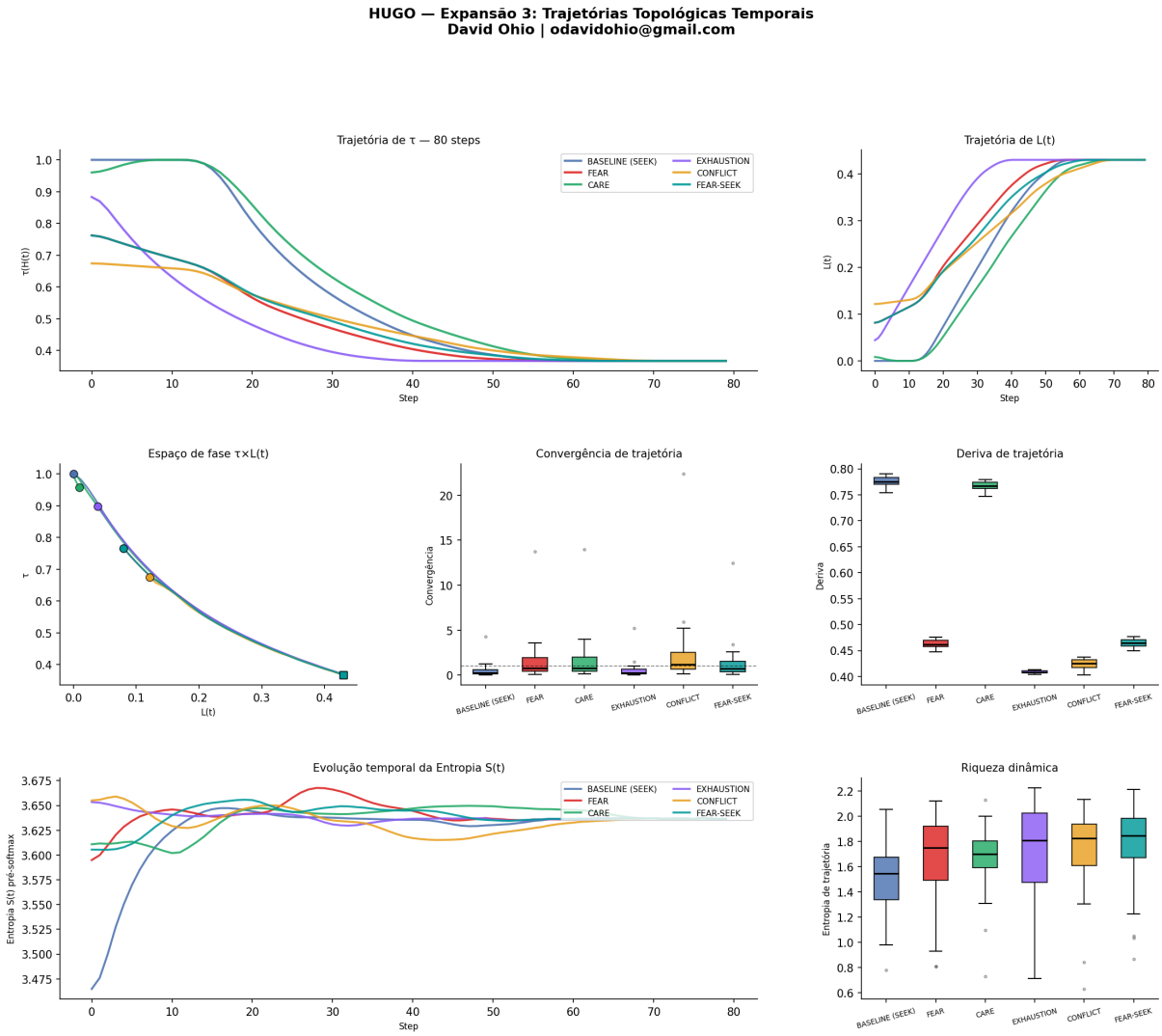

Figure 4: Expansion 3 unified panel. Top row: Mean trajectory of $\tau(H(t))$ over 80 steps (smoothed); mean trajectory of $L(t)$. Middle row: Phase space $\tau \times L(t)$; convergence distributions (boxplot); drift distributions. Bottom row: Entropy $\mathcal{H}_S(t)$ temporal evolution; trajectory entropy (richness of dynamics).

The $\tau(H(t))$ trajectories (top left) reveal a universal dissipative dynamic: all configurations exhibit declining temperature over 80 steps, converging toward $\tau(H(t)) \approx 0.35$–0.40 regardless of initial state. This is not a failure of architecture but a structural property of the homeostatic field: prolonged processing depletes homeostatic reserves, pushing $L(t)$ upward and $\tau(H(t))$ downward. The configurations differ in the shape of this dissipation.

6.3.2 Phase Space and the Universal Attractor

Figure 5 presents the phase portrait in $\tau \times L(t)$ space.

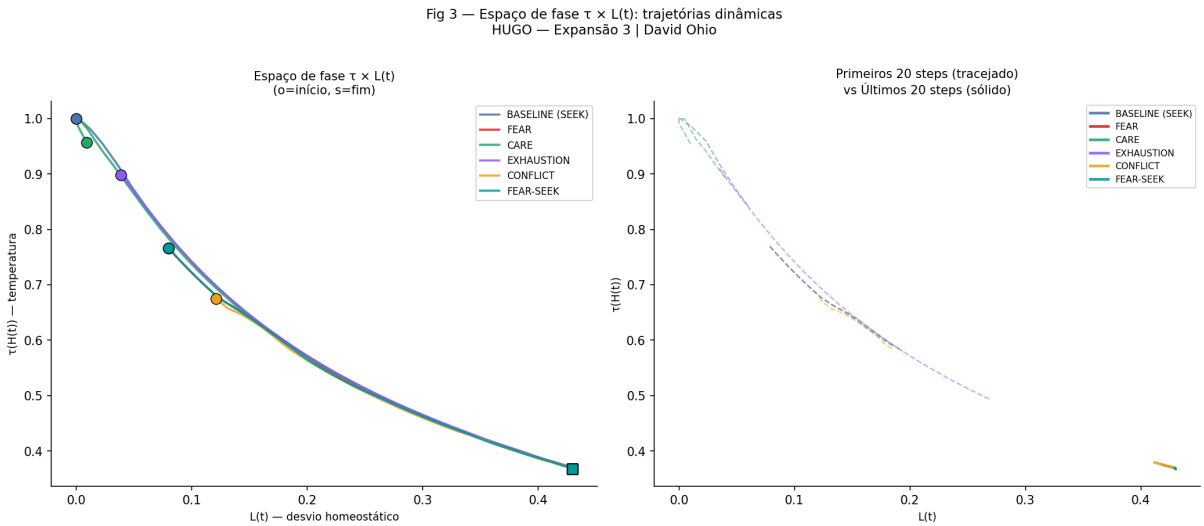

Figure 5: Expansion 3 — Phase space $\tau \times L(t)$. Left: Full 80-step trajectories. Circles mark initial states; squares mark final states. All configurations converge toward the same terminal region ($\tau \approx 0.35$–0.40, $L(t) \approx 0.40$–0.45). What differentiates configurations is the path, not the destination. Right: First 20 steps (dashed) vs. last 20 steps (solid), revealing that the final trajectories are nearly identical — the emotional "identity" is encoded in the initial dynamics, not the terminal state. This suggests that primary emotional signatures are transient structural features, not permanent attractors — consistent with biological affect which is inherently dynamic.

6.3.3 Convergence and Drift

Figure 6 presents the trajectory convergence analysis.

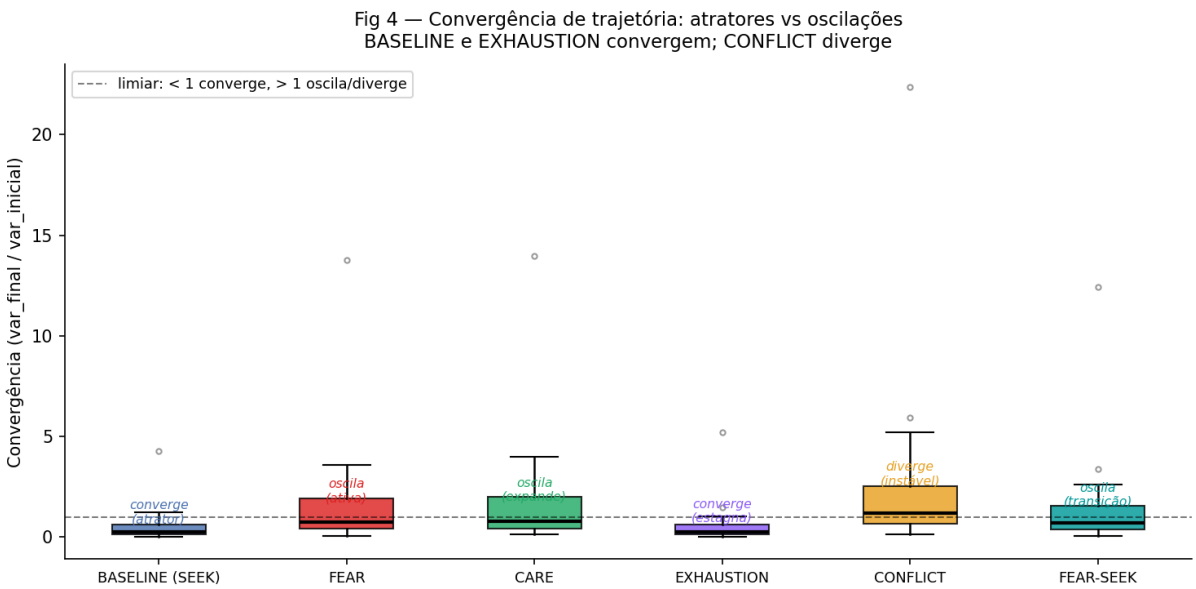

Figure 6: Expansion 3 — Trajectory convergence. Convergence = $\mathrm{Var}_{\mathrm{final}}/\mathrm{Var}_{\mathrm{initial}}$. Values < 1 indicate convergence to an attractor; values > 1 indicate divergence or oscillation. BASELINE and EXHAUSTION converge (mean ≈ 0.50); FEAR, CARE, and FEAR-SEEK oscillate (mean ≈ 1.4–1.6); CONFLICT strongly diverges (mean ≈ 2.4, maximum ≈ 22). The dashed line at 1.0 marks the convergence threshold. Notably, EXHAUSTION converges because it collapses — the variance of a stagnant system is low. BASELINE converges because it finds a processing attractor. These are topologically distinct convergence mechanisms producing the same scalar value.

The drift metric separates the configurations most cleanly: BASELINE (0.776) and CARE (0.767) exhibit ∼1.7× higher drift than EXHAUSTION (0.409) and CONFLICT (0.424). This reflects the distinction between exploratory processing (BASELINE, CARE), which traverses a large region of state space, and trapped processing (EXHAUSTION, CONFLICT, FEAR), which remains in a restricted attractor basin.

Statistical analysis confirmed significant differences for five of six trajectory metrics (Kruskal-Wallis, Table 6).

Table 6: Expansion 3: Kruskal-Wallis tests on trajectory metrics.

Metric	H statistic	p-value	Significant?
Convergence	36.13	8.9×10^{-7}	✓
Velocity ratio	3.22	0.667	—
Trajectory length	13.21	0.022	✓
Stability	144.81	$< 10^{-28}$	✓
Drift	164.44	$< 10^{-32}$	✓
Traj. entropy	17.46	3.7×10^{-3}	✓

The non-significance of velocity ratio is a genuine negative result: all configurations decelerate at comparable rates, reflecting the universal dissipative dynamic identified above. The rate of deceleration is not emotionally differentiated; the structure of the trajectory is.

7 Discussion

7.1 The Pre-Softmax Affective Intensity Principle: Empirical Validation

A central methodological commitment of HUGO is the measurement of topological features before softmax normalization, in addition to after it. We formalized the theoretical motivation in §3.4; the experimental results provide independent empirical validation.

Across all three experimental series, the post-softmax entropy $\mathcal{H}_A$ and pre-softmax entropy $\mathcal{H}_S$ exhibit systematic dissociation — they do not move together. Most strikingly, EXHAUSTION produces the minimum $\mathcal{H}_A$ among all configurations yet is not the minimum for $\mathcal{H}_S$. CONFLICT produces the minimum $\mathcal{H}_A$ (lowest post-softmax entropy, highest attentional concentration) despite having an intermediate $L(t)$ — less total homeostatic pressure than FEAR_EXTREME, yet more concentrated attention. This dissociation cannot be observed if one monitors only $A(t)$.

The implication for interpretability research is significant. Standard attention visualization tools operate exclusively on $A(t)$ (the post-softmax distribution), which is the input to downstream computations and the natural locus of mechanistic analysis. But $A(t)$ is not where affective pressure lives — it is where affective pressure arrives after compression. The structural magnitude of homeostatic gradients, the topology of the drive toward survival-relevant targets, the geometry of motivational conflict — all of these are encoded with higher fidelity in $S(t)$.

This is not a criticism of attention visualization as a technique for understanding what a network attends to. It is an argument that understanding why (in the sense of motivational causation) requires access to the pre-normalization field. HUGO is, to our knowledge, the first framework to formalize this distinction as an architectural principle and to provide empirical evidence for its importance through systematic dissociation analysis.

7.2 Emotions are Geometric, Not Categorical

The most consistent finding across three experimental series is that emotional analogs produce continuous topological gradients, not discrete categories. The FEAR gradient (Expansion 2)

demonstrates this most directly: Spearman $r = \pm 1.0$ for three of four metrics means that the topological geometry of the system tracks the intensity of the homeostatic perturbation in a perfectly monotone manner. There is no threshold below which FEAR "turns on." There is only a field — a geometry of constraint on information flow — that becomes progressively more restrictive as H_2 declines.

This is consistent with the dimensional view of affect [Russell, 1980, Barrett, 2017] and provides a computational mechanism: emotion is the shape of the constraint field on information flow, not a label applied to a discrete state.

7.3 The Universal Dissipative Attractor

The phase portrait of Expansion 3 reveals an unexpected property: all emotional configurations converge to the same terminal region after sufficient processing. This is a structural property of the homeostatic field — sustained processing depletes homeostatic reserves, driving the system toward the "exhaustion" region of phase space.

We interpret this as the computational analog of affect regulation through processing: the system transforms its emotional state through the act of cognition. The emotional "identity" is concentrated in the first 20–30 steps (initial dynamics), after which the system enters a universal regime. This is consistent with psychological models of emotional processing in which acute affect is processed toward a baseline state through continued cognitive engagement.

7.4 Correspondence with Classical Psychodynamic Constructs

The mathematical structure of HUGO maps onto psychoanalytic constructs in ways that are structurally legible, if not conclusive (Table 7).

Table 7: Correspondence between HUGO mathematics and psychoanalytic constructs.

Psychoanalytic concept	HUGO analog	Mathematical structure
Id (drives, instincts)	$H(t)$ — homeostatic pressures	Differential field
Ego (regulatory apparatus)	Network + $G(H(t))$ + $R(t)$	Functional composition
Unconscious	$R_{\text{perm}}(t)$ — permanent traces	Persistent deformations
Trauma	$\|R_{\text{perm}}\| > \theta_{\text{trauma}}$	Threshold exceedance
Transference	Resonance: R_{perm} on new input	Topological resonance
Elaboration / therapy	Controlled reduction of R_{perm}	Differential equation
Repetition compulsion	R_{perm} auto-activation	Memory attractor

The significance of this correspondence is not that HUGO "proves" psychoanalysis, nor that psychoanalysis validates HUGO. The point is narrower and more defensible: the mathematics independently generates structures that share formal properties with constructs developed from clinical observation. Whether this convergence reflects deep structural necessity or productive coincidence is an open question. What it suggests, at minimum, is that the HUGO formalism is not architecturally arbitrary — it recapitulates, from first principles, distinctions that a century of clinical practice found it necessary to make.

7.5 Implications for AGI Architecture

The results suggest that affective analogs are not optional "personality features" to be added to cognitive systems post hoc, but necessary structural properties of any system that must regulate its own information processing under resource constraints.

A system without homeostatic regulation has no intrinsic mechanism for determining when to focus vs. explore, when to persist vs. disengage, when to engage socially vs. avoid. The HUGO

framework provides a tractable, mathematically grounded implementation of these regulatory functions through the homeostatic temperature $\tau(H(t))$ and the gradient modulator $G_{\text{total}}(t)$.

Critically, the results here are obtained with $R(t) = 0$ (no memory). The introduction of persistent topological memory ($R_{\text{perm}} \neq 0$) is expected to produce qualitatively richer dynamics — including the possibility of individual "perspectives" that diverge between instances with identical architectures but different histories. This will be the subject of Paper 3 in the HUGO series.

7.6 Limitations

The present work has several important limitations.

Artificial inputs. All experiments use Gaussian random inputs. The topological signatures reported here are properties of the homeostatic-attentional system, not of semantic content. Whether these signatures persist, amplify, or transform when processing structured semantic inputs remains an open question.

Architecture specificity. Results are obtained with a specific Gray-Box Structural Attention architecture. Generalization to other attention mechanisms (e.g., linear attention, flash attention) requires further investigation.

Biological grounding. The correspondence to biological affect is proposed as a structural analogy, not a physiological identity. We do not claim that the Wasserstein distance of synthetic attention matrices measures the same quantities as cortical activity. The claim is structural: the mathematical form is homologous.

8 Conclusion

We have demonstrated that homeostatic field configurations produce statistically distinguishable, structurally interpretable topological signatures in neural information flow. Three experimental series establish: (i) primary emotional analogs are separable with $p < 10^{-8}$ via six topological metrics; (ii) FEAR is a continuous topological field, not a discrete category; (iii) distinct trajectory topologies characterize exploratory, convergent, and divergent emotional regimes over time.

These results constitute the experimental foundation of the HUGO framework (Homological Unified Gradient Ontology), which advances the hypothesis that affective regulation and cognitive organization admit a shared topological description. The psychoanalytic correspondences identified in Section 7 are offered as structural observations — independently derived mathematical forms that converge on constructs developed from clinical experience — not as the framework's central contribution. The central contribution is mechanistic: homeostatic gradients produce measurable, continuous topological geometry in neural information flow.

The present results establish that primary affect-like regimes can be generated, measured, and topologically discriminated within a single homeostatically regulated network. This raises an immediate extension: whether such regimes can be altered not only by internal disequilibrium, but by the inferred emotional state of another system communicated through language. If so, affect would no longer be merely an internal regulatory geometry — it would be the basis of structured inter-system coupling. That is the question addressed by the ECHO framework (Empathic Coupling via Homeostatic Oscillation, Paper 2 of this series), which introduces a language-mediated perturbation layer on the homeostatic field established here. Without the present foundation, ECHO would have no formal ground. With it, the coupling hypothesis becomes testable.

Future work will further introduce persistent topological memory ($R_{\text{perm}}(t)$, Paper 3) and the application of the Structural Radiant [Ohio, 2025c] to the trajectory data generated here.

Data and Code Availability. All experimental code, data, and figure generation scripts are available in the HUGO GitHub repository. Results data archived at Zenodo (DOI: 10.5281/zenodo.XXXXXXX). License: Creative Commons Attribution 4.0 International (CC BY 4.0).

Conflict of Interest. None declared.

Acknowledgments. This work was conducted independently. The name HUGO honors Hugo, whose existence is the primary motivational field from which this research emerges.

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